

Memo: Where We Stand on `/kb-regenerate`

Date: 2026-04-29 **Author:** Black (with Claude) **Audience:** EAGLE team **TL;DR:** The `/kb-regenerate` command exists, works, and is the only thing keeping our KB metadata, checklists, templates, and compliance matrix from silently drifting. It is **not wired into CI** — it only runs when a human invokes the slash command in Claude Code. Given the churn in the KB the last 4 weeks (v3 path migration, intake-gate matrix keys, S3 Vectors lane), we should (a) run it now against `dev`, and (b) decide whether to promote it from a manual hygiene tool to a scheduled or pre-deploy check.

What it is

A Claude Code slash command at `.claude/commands/kb-regenerate.md`, introduced in commit **a522b2d** (2026-04-03) and last updated in **eff6b45** (2026-04-27, Alvee's KB v3 path fix). It runs up to five phases:

Phase	Purpose	What it touches
1. Orphan purge	Scan <code>eagle-document-metadata-dev</code> , head-check each <code>s3_key</code> , delete entries whose S3 object is gone	DynamoDB
2. Checklist + template validation	<code>head_object</code> every checklist (9) and template (11) the backend hardcodes against	S3 only — read
3. Compliance matrix deep analysis	<code>scripts/kb_matrix_analysis.py</code> — three layers: <code>matrix.json</code> internal consistency, backend <code>compliance_matrix.py</code> sync, frontend <code>matrix-data.ts</code> sync (Ctrl+M modal)	Source files only — read
4. Metadata regeneration	Optional, behind <code>--regenerate <key></code> or <code>--regenerate-all</code> . Invokes <code>eagle-metadata-extractor-dev</code> Lambda asynchronously per S3 object	Lambda + DynamoDB
5. Summary report	Consolidated PASS/FAIL table	stdout only

Default invocation runs Phases 1–3 + 5. Phase 4 is opt-in.

How to invoke

From inside Claude Code (any session in this repo):

```
/kb-regenerate                # full hygiene run (dry orphan delete OFF – will p
/kb-regenerate --dry-run      # report only, deletes nothing
/kb-regenerate --regenerate <s3-key> # re-extract metadata for one document
/kb-regenerate --regenerate-all  # re-extract metadata for every doc under approved
```

It needs the `eagle` AWS SSO profile (us-east-1, account 695681773636 / NONPROD). The command body uses inline `python -c "..."` blocks invoked via Bash, so any environment that can run `boto3` against the dev account works.

Is it wired up?

No — not in the way "wired up" usually means.

- ❌ Not in `.github/workflows/` — no scheduled or PR-triggered run.
- ❌ Not in `Justfile` — no `just kb-regenerate` recipe.
- ❌ Not in `server/app/` — no API endpoint.
- ❌ Not in the deploy pipeline — `deploy.yml` does not gate on KB integrity.
- ✅ Referenced in two docs: the KB v2 migration plan (`docs/20260413-100000-plan-kb-v2-migration-v1.md` , Phase 4 step 8) and the weekly changelog. Both treat it as a manual cleanup step.
- ✅ The Phase 3 helper (`scripts/kb_matrix_analysis.py`) is a standalone Python script — runnable without Claude.

In practice that means the only way it executes today is when a developer remembers to type `/kb-regenerate` in their Claude session. That is fine for the orphan-purge phase (destructive, wants a human), but Phases 2 and 3 are pure validators and should arguably run on every PR that touches `eagle-plugin/data/matrix.json` , `compliance_matrix.py` , `template_registry.py` , `knowledge_tools.py` , or `matrix-data.ts` .

Recent commits that bear on whether the command needs enhancement

Commit	Date	Why it matters
a522b2d	2026-04-03	Created <code>/kb-regenerate</code> and <code>kb_matrix_analysis.py</code> . Established the 5-phase shape.
af83106	2026-04-02	Compliance matrix Phases 2e–6 landed: <code>_CONFIDENCE</code> , <code>_extract_far_citations</code> , <code>_verify</code> , <code>get_template_fields</code> , <code>_related_far</code> . Phase 3 of kb-regenerate does not yet check confidence-dict consistency or <code>_verify</code> output shape.

Commit	Date	Why it matters
e39b567	2026-04-09 (est.)	Added new matrix keys: <code>intake_required_facts</code> , <code>budget_semantics</code> . Phase 3 layer-A does not yet validate these new top-level keys.
e12fcdd / d14a41d	mid-Apr	Tier compliance docs (core vs supplemental). New schema fields under <code>doc_rules</code> . Layer A check should grow accordingly.
7ebbf7a	mid-Apr	Semantic retrieval Lane 1e (S3 Vectors + Titan v2). Nothing in kb-regenerate validates vector index sync against <code>eagle-documents</code> bucket — when a doc is deleted from S3, its vector lingers.
eff6b45	2026-04-27	KB v3 path migration. Hardcoded paths in kb-regenerate's CHECKLISTS / TEMPLATES lists had to be hand-edited in lockstep with <code>knowledge_tools.py:BUILTIN_KB_ENTRIES</code> . This is a maintenance smell — both lists are duplicates of the same source of truth.
cf0d3bb / 8213928	2026-04-26 → 2026-04-28	Per-lane source transparency + lane breakdown modal. Adds new SSE wire-shape (<code>lane</code> , <code>score</code> , <code>lane_breakdown</code>). Surface area where matrix→frontend drift would now be user-visible.

Suggested enhancements (rough priority order)

- De-duplicate the checklist + template lists.** Phase 2 hardcodes 9 checklists and 11 templates. The same lists live in `server/app/tools/knowledge_tools.py` (`BUILTIN_KB_ENTRIES`) and `server/app/template_registry.py` (`TEMPLATE_REGISTRY`). Have Phase 2 import from those modules instead. KB v3 only had to touch four files because of this duplication; next migration shouldn't.
- Teach Phase 3 about new matrix keys.** Add Layer-A validators for `intake_required_facts` and `budget_semantics` (commit e39b567), and core-vs-supplemental tier fields (commit e12fcdd).
- Add a Phase 3D: `_CONFIDENCE` and `_verify shape check`.** Confirm every `doc/compliance/approval` entry in `get_requirements()` output carries a `confidence` field (Phase 6 from the matrix overhaul) and that `_verify.far_citations` is non-empty when applicable. Could be a 30-line property-style assertion.
- Vector index sync (Phase 6 — new).** When a doc is deleted from S3, Phase 1 purges DynamoDB metadata but leaves the S3 Vectors index entry. Add a Phase 6 that diffs vector IDs against current S3 keys.
- Wire Phases 2 + 3 into CI.** They are read-only and fast. Make a `just kb-validate` recipe that runs `python scripts/kb_matrix_analysis.py` plus an extracted version of Phase 2, and add a GitHub Action that runs on PRs touching the trigger files. Phases 1 and 4 stay manual.
- Add a `--qa` flag.** Hardcoded constants reference `eagle-document-metadata-dev` and `eagle-documents-695681773636-dev` . QA and prod accounts will need their own values; today the command can only target dev.

Recommended next action

Run `/kb-regenerate --dry-run` against dev today to get a baseline orphan count and surface any KB v3 path issues we missed. Decide afterward whether enhancement #5 (CI gating for Phases 2 + 3) is worth a phase plan.

Files referenced in this memo

Path	Why
<code>.claude/commands/kb-regenerate.md</code>	The command itself
<code>scripts/kb_matrix_analysis.py</code>	Phase 3 helper
<code>server/app/compliance_matrix.py</code>	Backend matrix code (Layer B in Phase 3)
<code>server/app/tools/knowledge_tools.py</code>	<code>BUILTIN_KB_ENTRIES</code> source of truth for checklists
<code>server/app/template_registry.py</code>	<code>TEMPLATE_REGISTRY</code> source of truth for templates
<code>eagle-plugin/data/matrix.json</code>	Compliance matrix source of truth (Layer A in Phase 3)
<code>client/components/contract-matrix/matrix-data.ts</code>	Frontend mirror (Layer C in Phase 3)
<code>docs/20260413-100000-plan-kb-v2-migration-v1.md</code>	Earlier migration plan that calls <code>/kb-regenerate</code> as Step 8